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* What is DNS?

- DNS (Domain Name System) is the internet's phone book. Humans remember names like google.com but computers communicate using IP addresses like 142.250.195.42. DNS translates one to the other.

* DNS Record Types

Record	Full Name	Purpose
A	Address	Domain → IPv4 address
AAAA	IPv6 Address	Domain → IPv6 address
MX	Mail Exchange	Where to send emails
CNAME	Canonical Name	Alias → another domain
NS	Name Server	Which server handles DNS for domain
TXT	Text	Verification, SPF records
PTR	Pointer	Reverse DNS (IP → domain)

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* How DNS Works

- When you type google.com and press enter:

1) Your browser checks its local cache — already known this IP?

2) If not, asks your OS resolver (checks hosts file).

3) If not, asks your Recursive Resolver (usually your router / IPS — 192.168.0.1).

4) Resolver asks a Root Name Server → "who handles .com?"

5) Root server replies → TLD Name Server (handles .com).

6) TLD server replies → Google's Authoritative Name Server.

7) Authoritative server replies with the actual IP address

8) Resolver caches the answer and sends it back to you or your machine.

9) Then your browser actually connects to that IP address.

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* Recursive vs Iterative Queries

Type	Who does the work	Used by
Recursive	Resolver does all the work for you	Your PC → Resolver
Iterative	Each server just points to the next	Resolver → Root/TLD/Authoritative

* DNS in Wireshark — what to look for

Field	What it means
Transaction ID	Matches query to response (eg: 0x8642).
Query Type "A"	Asking for IPv4 address.
Query Name	The domain being looked up.
Response IP	The answer — IP address returned
TTL	Time To Live — how long to cache this answer

* DNS and DFIR — Why This Matters

- Every malware infection generates DNS queries — malware must find its C2 server by domain name.

- DGA (Domain Generation Algorithm) — malware generates random-looking domains like xkq7z3mpa.ru — these look nothing like real domains.

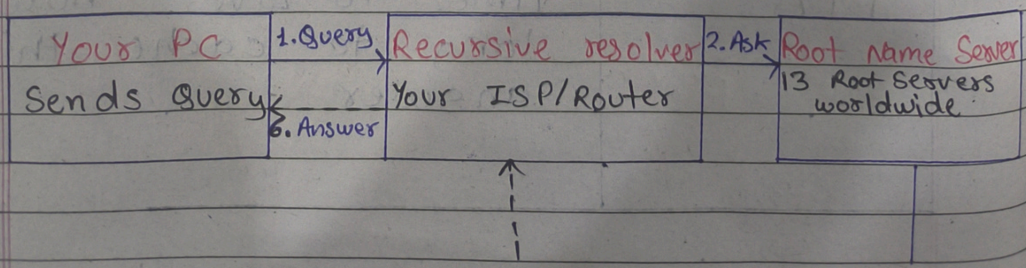
- DNS tunneling — attackers can hide data inside DNS queries to exfiltrate information.

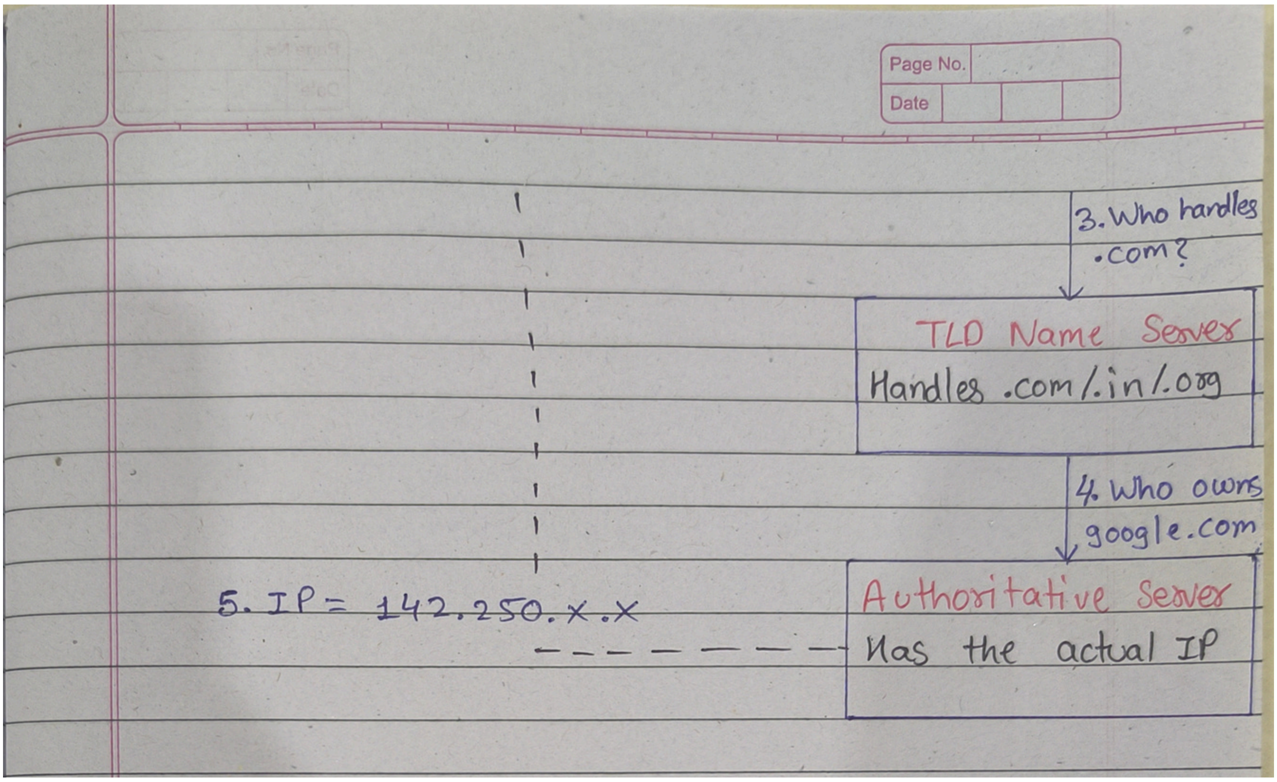
- High DNS query volume to one domain = beaconing

- The filter dns in Wireshark shows everything your machine is talking to — even encrypted traffic reveals destination domains via DNS.

* DNS Resolution Flow

- What happens when you type google.com:

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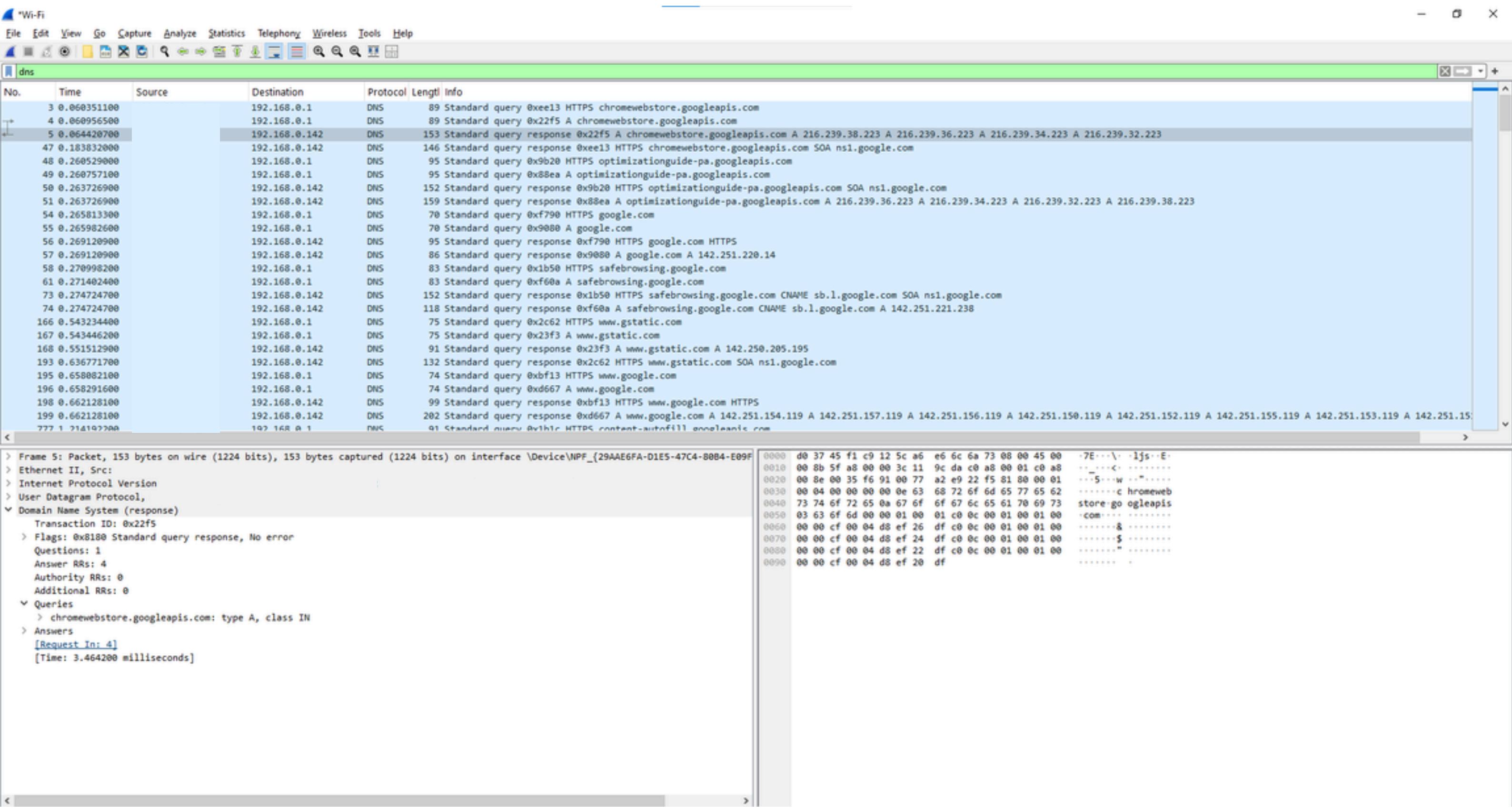
3. Who handles .com?
TLD Name Server
Handles .com/.in/.org
4. Who owns google.com
Authoritative Server
Has the actual IP
5. IP = 142.250.x.x


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Lengtl Info
3 0.060351100 192.168.0.1 DNS 89 Standard query 0xee13 HTTPS chromewebstore.googleapis.com
4 0.060956500 192.168.0.1 DNS 89 Standard query 0x22f5 A chromewebstore.googleapis.com
5 0.064420700 192.168.0.142 DNS 153 Standard query response 0x22f5 A chromewebstore.googleapis.com A 216.239.38.223 A 216.239.36.223 A 216.239.34.223 A 216.239.32.223
47 0.183832000 192.168.0.142 DNS 146 Standard query response 0xee13 HTTPS chromewebstore.googleapis.com SOA ns1.google.com
48 0.260529000 192.168.0.1 DNS 95 Standard query 0x9b20 HTTPS optimizationguide-pa.googleapis.com
49 0.260757100 192.168.0.1 DNS 95 Standard query 0x88ea A optimizationguide-pa.googleapis.com
50 0.263726900 192.168.0.142 DNS 152 Standard query response 0x9b20 HTTPS optimizationguide-pa.googleapis.com SOA ns1.google.com
51 0.263726900 192.168.0.142 DNS 159 Standard query response 0x88ea A optimizationguide-pa.googleapis.com A 216.239.36.223 A 216.239.34.223 A 216.239.32.223 A 216.239.38.223
54 0.265813300 192.168.0.1 DNS 70 Standard query 0xf790 HTTPS google.com
55 0.265982600 192.168.0.1 DNS 70 Standard query 0x9080 A google.com
56 0.269120900 192.168.0.142 DNS 95 Standard query response 0xf790 HTTPS google.com HTTPS
57 0.269120900 192.168.0.142 DNS 86 Standard query response 0x9080 A google.com A 142.251.220.14
58 0.270998200 192.168.0.1 DNS 83 Standard query 0x1b50 HTTPS safebrowsing.google.com
61 0.271402400 192.168.0.1 DNS 83 Standard query 0xf60a A safebrowsing.google.com
73 0.274724700 192.168.0.142 DNS 152 Standard query response 0x1b50 HTTPS safebrowsing.google.com CNAME sb.l.google.com SOA ns1.google.com
74 0.274724700 192.168.0.142 DNS 118 Standard query response 0xf60a A safebrowsing.google.com CNAME sb.l.google.com A 142.251.221.238
166 0.543234400 192.168.0.1 DNS 75 Standard query 0x2c62 HTTPS www.gstatic.com
167 0.543446200 192.168.0.1 DNS 75 Standard query 0x23f3 A www.gstatic.com
168 0.551512900 192.168.0.142 DNS 91 Standard query response 0x23f3 A www.gstatic.com A 142.250.205.195
193 0.636771700 192.168.0.142 DNS 132 Standard query response 0x2c62 HTTPS www.gstatic.com SOA ns1.google.com
195 0.658082100 192.168.0.1 DNS 74 Standard query 0xbf13 HTTPS www.google.com
196 0.658291600 192.168.0.1 DNS 74 Standard query 0xd667 A www.google.com
198 0.662128100 192.168.0.142 DNS 99 Standard query response 0xbf13 HTTPS www.google.com HTTPS
199 0.662128100 192.168.0.142 DNS 202 Standard query response 0xd667 A www.google.com A 142.251.154.119 A 142.251.157.119 A 142.251.156.119 A 142.251.150.119 A 142.251.152.119 A 142.251.155.119 A 142.251.153.119 A 142.251.15
Frame 5: Packet, 153 bytes on wire (1224 bits), 153 bytes captured (1224 bits) on interface \Device\NPF_{29AAE6FA-D1E5-47C4-80B4-E09F
Ethernet II, Src:
Internet Protocol Version
User Datagram Protocol,
Domain Name System (response)
Transaction ID: 0x22f5
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 4
Authority RRs: 0
Additional RRs: 0
Queries
chromewebstore.googleapis.com: type A, class IN
Answers
[Request In: 4]
[Time: 3.464200 milliseconds]